

MUHAMMED RAFI K P

MERN Stack Developer

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 [LinkedIn](#) |  [GitHub](#) |  [LeetCode](#)

PROFILE SUMMARY

Skilled Full Stack Developer specializing in scalable web applications. Experienced in both frontend and backend development with strong problem-solving abilities. Passionate about writing clean code, optimizing performance, and delivering user-centric solutions.

TECHNICAL SKILLS

- **Programming Languages:** JavaScript, TypeScript, Java (Basics), C (Basics)
- **Frontend Technologies:** HTML5, CSS3, React.js, Redux, Tailwind CSS, Bootstrap, Material-UI
- **Backend Technologies:** Node.js, Express.js, WebSocket, RESTful APIs, Redis
- **Databases & Cloud Storage:** MongoDB, Firebase, PostgreSQL, Amazon S3, Cloudinary
- **DevOps & Deployment:** Docker, Kubernetes, Git, GitHub, GitHub Actions, CI/CD Pipelines
- **Microservices Architecture:** RabbitMQ, gRPC, REST-based service communication, Design Patterns
- **Tools:** Postman, Figma

PROJECTS

• Greenora - Smart Waste Management Application

 [GitHub](#) |  [Live Link](#)

An end-to-end microservices-based smart waste management solution enabling users to request waste and scrap pickups, track waste collection in real-time, and promote sustainable practices through intelligent features and seamless communication.

- Designed a responsive, user-friendly UI using **React.js**, **Redux Toolkit**, and **Tailwind CSS**.
- Secured authentication with **JWT**, **Google OAuth**, and added token blacklisting to prevent unauthorized access.
- Implemented real-time chat and notifications with **Socket.IO** for instant communication.
- Integrated **Razorpay** payment gateway for seamless transaction processing.
- Built a **chatbot** using **Gemini API** to assist users with waste management queries.
- Integrated location services and **Mapbox API** for accurate location tracking.
- Used **Redis** caching to optimize application performance.
- Leveraged **RabbitMQ** for asynchronous messaging and **gRPC** for synchronous communication between microservices.
- Deployed on **Google Kubernetes Engine (GKE)** within **Google Cloud Platform (GCP)** for scalability and reliability.
- Leveraged **AWS S3** for secure cloud storage of user data.

Tech Stack: React.js, Redux Toolkit, Tailwind CSS, Node.js, Express.js, MongoDB, RabbitMQ, gRPC, Redis, Socket.IO, Razorpay, Mapbox, Google OAuth, AWS S3, GCP (GKE), Vercel, JWT

• Stepex - E-commerce Application

 [GitHub](#) |  [Live Link](#)

Built a dynamic and scalable e-commerce platform where users can browse, explore, and purchase shoes from a wide variety of top brands.

- Designed an intuitive and visually appealing UI to enhance the user shopping experience.

- Built a real-time shopping cart system supporting seamless product selection, quantity updates, and pricing, boosting conversion rates.
- Implemented advanced product filtering, search, and sorting features to help users quickly find their ideal shoes.
- Integrated **Razorpay** for secure and reliable payment processing.
- Built an intuitive admin panel for efficient product, order, and user management.
- Added promotional features including offers, coupons, referral programs, and a built-in wallet system to increase user engagement and retention.
- Deployed the application using **PM2** and **Nginx** on **AWS** for reliable performance and scalability.

Tech Stack: Node.js, Express.js, MongoDB, EJS, Bootstrap, CSS, Razorpay, GitHub, AWS, PM2, MongoDB Atlas, Nginx.

MINI PROJECTS

• Aadhaar OCR System

 [GitHub](#) |  [Live Link](#)

Developed a client-server application to extract structured Aadhaar card data from images using Google Cloud Vision API and intelligent parsing logic.

- Implemented secure image upload and backend integration using **Node.js**, **Express**, and **Multer**.
- Utilized **Google Cloud Vision API** for OCR and built custom parsing logic using **RegEx** for accurate field extraction.
- Displayed structured data (name, Aadhaar number, DOB, gender, address, pincode) in a clean and responsive React-based UI.
- Handled noisy OCR outputs gracefully and ensured robust parsing of scanned Aadhaar cards.

Tech Stack: React.js, TypeScript, Node.js, Express.js, Tailwind CSS, Google Cloud Vision API, Vite, Multer, Axios.

• User Management System

 [GitHub](#)

Developed a secure user management system featuring authentication, role-based access control, and profile management.

- Deployed secure login using **JWT** for authentication.
- Engineered role-based access control to manage user permissions effectively.
- Enabled user profile creation, editing, and management through a **React.js** frontend with **Redux** state management.

Tech Stack: Node.js, Express.js, MongoDB, JWT, React.js, Redux.

• React Netflix Clone

 [GitHub](#)

Built a responsive Netflix clone that features dynamic movie browsing and secure user authentication.

- Developed an interactive UI to browse movies, TV shows, and watch trailers seamlessly.
- Leveraged **TMDB API** for real-time movie and show data.
- Implemented **Firebase** authentication to manage user profiles and personalized watchlists.
- Optimized performance using lazy loading and ensured responsive design across devices.

Tech Stack: React.js, Firebase, Tailwind CSS, TMDB API.

EXPERIENCE

- **Ecomex**  Sep 2024 – Present
Freelance Web Developer Remote
 - Created modern, responsive portfolio websites tailored to small business needs, improving their online presence.
 - Crafted mobile-first UI using **HTML5**, **CSS3**, and **JavaScript** frameworks for better user engagement.
 - Collaborated with clients to deliver tailored solutions.

EDUCATION

- **MERN Stack Development** Nov 2023 - Present
Prototype, Calicut
- **Bachelor of Science in Physics** 2020 - 2023
University of Calicut